

Scientific-contract audit of the gamma-gated publication campaign

exp083 · 12 August 2026 · Draft

Abstract

We audited the scientific contract of the gamma-gated sparsity campaign before production compute. The review followed exp022's 90 registered training cells and all 17 collection experiments through planning, model construction, checkpoint selection, inference, aggregation, provenance, and publication. The dry production plan created no scheduler jobs. The audit found 2 blockers, 5 important findings, and 1 minor finding. The campaign is not ready for production: checkpoint epochs can differ from the epochs claimed in the writings, and cell validation can accept mismatches in scientifically central parameters. Focused issues record each required implementation change.

Methods

1. **Freeze the reviewed source.** We audited commit `339c164c3e553b5888e36be30a539ff566cadcd0d`. The checkout was clean when the campaign manifest was generated, and the manifest records the same commit and the frozen lockfile hash.
2. **Generate the production plan without submission.** We initialized a production-profile campaign in an external disposable directory and ran the collection `submit` command without `--live` or `--test-only`. This resolved cell lists, commands, dependencies, resources, and expected outputs but did not call `sbatch`. Private paths and resource identities were replaced with publication placeholders before the plan was retained.
3. **Trace exp022 parameters.** For every cell, we followed the registry through `build_train_args`, the SNN command line, model and training construction, `config.json`, `metrics.json`, checkpoint files, campaign validation, and exp022's writing. We compared argument values within and across the six training-run families and checked that architecture-specific differences were registered.
4. **Trace downstream contracts.** For each collection node, we reviewed declared dependencies, runner arguments, training checkpoint identity, inference mutations, sample selection, random seeds, cache behavior, aggregation unit, uncertainty summary, required outputs, collection provenance, and the corresponding writing.
5. **Classify and disposition concerns.** A blocker can make production scientifically wrong, irreproducible, or materially misreported. An important finding is a meaningful ambiguity or provenance gap that must be resolved or explicitly documented. A minor finding cannot plausibly change a conclusion. A false alarm is an investigated contract

that agrees across source, artifacts, and prose. Material implementation defects were not fixed in this audit; each received a focused issue with an acceptance test.

Exact production plan

The dry plan contains 90 training cells and 23 scheduler job definitions. It creates five exp022 arrays, one aggregation job, sixteen downstream jobs, and one finalization job. The command produced 0 paid-compute or scheduler jobs.

Training run	Cells	Role
TR-01	6	Canonical COBA and PING reference
TR-02	36	Spike-budget sweep
TR-03	18	Inhibitory-timescale sweep
TR-04	15	Integration-timestep sweep
TR-05	12	Recurrent-initialization sweep
TR-06	3	Variable-rate streaming bank

Resource tier	Cells	Scientific shape
Standard	78	Sweep cells at the standard timestep
Fine timestep	3	TR-04 cells at the smallest timestep
Canonical COBA	3	Full pooled-MNIST COBA cells
Canonical PING	3	Full pooled-MNIST PING cells
Variable rate	3	TR-06 output-LIF cells

The preserved `production-plan.json` contains the complete sanitized collection plan, all exp022 cell commands, and all dry submission commands. Its campaign source is `339c164c3e553b5888e36be30a539ff566cadcd0d`. The recorded exp022 manifest and resource-file SHA-256 prefixes are `c9551a5f19b1` and `ec8e48adaf3f` the preserved plan contains both complete hashes.

Reviewed execution graph

Experiment	Dependencies	Training run	Scheduled arguments
exp022	root	none	none
exp023	root	none	none
exp047	root	none	none
exp080	root	none	none
exp081	root	none	none
exp024	exp022	TR-02	–skip-training
exp025	exp022	TR-02	–only-missing
exp037	exp022	TR-02	–skip-training
exp038	exp022	TR-02	–skip-training
exp041	exp022	TR-03	–skip-training
exp044	exp022	TR-04	–skip-training

Experiment	Dependencies	Training run	Scheduled arguments
exp049	exp022	TR-05	none
exp082	exp022	TR-06	none
exp033	exp041	none	none
exp042	exp041	none	–skip-training
exp046	exp041	none	none
exp054	exp041	none	none

Findings

Blockers

F01: Best-accuracy checkpoints are reported as final-epoch dynamics

Observed behavior. Training saves weights.pth at the best internal-holdout epoch and weights_final.pth at the last epoch. Headline downstream inference reloads weights.pth while methods or captions describe epoch-50 dynamics.

Expected scientific meaning. Every reported quantity identifies and loads the checkpoint epoch its prose claims.

Publication risk. Firing rates and recurrent weights can continue changing after accuracy plateaus, so the reported dynamical state can be from the wrong epoch.

Disposition. Issue #74 defines and tests explicit best-versus-final checkpoint roles and downstream checkpoint provenance. Tracked in #74.

Code evidence. tools/snn/train.py:926 (`train`); tools/snn/train.py:927 (`train`); experiments/exp041.py:156 (`\_infer\_cell`); writings/exp041.typ:76 (`Convergence`); experiments/exp044.py:127 (`\_infer\_cell`).

F02: Cell validation omits scientific manifest arguments

Observed behavior. The manifest records –w-in and the recurrent trainability flags, but ARG_TO_CONFIG maps none of them, so validate_cell never compares them with saved config.

Expected scientific meaning. Every scientific manifest argument is validated against its resolved saved value, or validation fails closed.

Publication risk. A wrong input distribution or wrong TR-05 trainability intervention can be accepted into the production bank.

Disposition. Issue #75 adds complete argument coverage and fail-closed tests. Tracked in #75.

Code evidence. experiments/exp022_support/campaign.py:209 (`ARG\_TO\_CONFIG`); experiments/exp022_support/campaign.py:233 (`\_expected\_config`); experiments/exp022.py:113 (`MODEL\_RECIPES`); experiments/exp022.py:261 (`\_init\_cells`).

Important findings

F03: exp025 discards two registered theta-u seeds

Observed behavior. TR-02 registers three seeds at every theta-u value, but exp025 returns only seed 42 for each regularized condition.

Expected scientific meaning. The published frontier aggregates all three registered cells per condition, with representative single-seed diagnostics labeled separately.

Publication risk. The primary frontier uses a different aggregation unit and less replication than exp022 promises.

Disposition. Issue #76 updates consumers, metadata, and aggregation tests. Tracked in #76.

Code evidence. experiments/exp022.py:188 (``seeds_for``); experiments/exp025.py:156 (``seeds_for``); writings/exp022.typ:141 (``TR-02 table``).

F04: exp044 reports the wrong training batch size

Observed behavior. TR-04 cells inherit `-batch-size 256` from exp022, while exp044 records and publishes batch size 64.

Expected scientific meaning. A consumer derives training metadata from the checkpoint manifest rather than a retired local training constant.

Publication risk. All 15 timestep-sweep cells have a materially misstated optimization method.

Disposition. Issue #77 derives and verifies exp044 training provenance from consumed cells. Tracked in #77.

Code evidence. experiments/exp022.py:120 (``MODEL_RECIPES``); experiments/exp044.py:63 (``BATCH_SIZE``); writings/exp044.typ:39 (``Methods``).

F05: exp023 machine-readable drive provenance contradicts execution

Observed behavior. The runner executes synthetic-spikes at 5 Hz for COBA and 45 Hz for PING, but numbers.json is assembled with an MNIST sample label and one 50 Hz value.

Expected scientific meaning. Saved provenance distinguishes each raster operating point and the f-I sweep grid.

Publication risk. The displayed untrained operating conditions cannot be reconstructed from the published data contract.

Disposition. Issue #78 makes saved drive provenance derive from executed arguments. Tracked in #78.

Code evidence. experiments/exp023.py:52 (``COBA_INPUT_RATE_HZ``); experiments/exp023.py:53 (``PING_INPUT_RATE_HZ``); experiments/exp023.py:890 (``main payload``); experiments/exp023.py:894 (``main payload``).

F06: The pooled MNIST holdout is called test data without a checkpoint-selection contract

Observed behavior. The loader concatenates the official 60,000-image training and 10,000-image test partitions, optionally subsamples, then creates a fixed stratified 80/20 split. That 20% split is evaluated every epoch and selects weights.pth.

Expected scientific meaning. Publication text calls this an internal pooled-MNIST holdout used for checkpoint selection, not an untouched or official test set, and reports 56,000/14,000 or 5,600/1,400 train/holdout counts.

Publication risk. Readers can interpret repeatedly consulted checkpoint-selection data as a final held-out test set and can read 70,000 or 7,000 as the number of optimizer-training images.

Disposition. Documented resolution: treat the split as an internal holdout throughout this campaign, state that the official partitions were pooled, and reserve final-test language for a future untouched evaluation. No scientific rerun is implied by this naming correction.

Code evidence. tools/snn/datasets.py:123 (``load_dataset``); tools/snn/datasets.py:158 (``load_dataset``); tools/snn/train.py:753 (``train``); writings/exp022.typ:121 (``TR-01 table``).

F08: The production manifest leaves scientific defaults unresolved

Observed behavior. Cell commands omit the fixed input rate, topology, Dale constraint, weight decay, gradient clip, and AMPA decay; those values are inherited from shared CLI or model defaults and are absent from the manifest's resolved parameter block.

Expected scientific meaning. The pre-run manifest states every scientific setting with physical meaning and units, including inherited values.

Publication risk. The frozen source preserves reproducibility, but the plan is not cold-readable and a shared default change can alter a regenerated campaign without a registry-level parameter diff.

Disposition. Issue #79 defines a complete resolved scientific schema and cross-cell difference check. Tracked in #79.

Code evidence. experiments/exp022.py:367 (``build_train_args``); tools/snn/train.py:207 (``train``); tools/snn/train.py:42 (``GRAD_CLIP``); writings/exp022.typ:122 (``TR-01 table``).

Minor finding

F07: Dry-run submission requires the configured uv executable

A dry submission invokes the configured uv executable to ask exp022 for retry-only cell lists; a placeholder executable prevents plan generation even though no scheduler submission is requested. Operational surprise only; no result or production job is changed. Accepted limitation for this campaign and recorded in the reproducer.

Code evidence. experiments/collections/gamma_gated_sparsity/slurm.py:123 (``_exp022_cells``).

Checked contracts that held

1. **COBA/PING matched initialization.** Both recipes use the same `-w-in`, sparsity, readout mode, readout initializer, learning rate, batch size, duration, timestep, and seeds. Registered differences are `-ei-strength` and `-v-grad-dampen`.
2. **Smoke/production separation.** Initialization records `profile=smoke` only when `-smoke` is supplied, uses `-plumbing` only for that profile, and requires a unique external campaign root.
3. **Source and cache identity.** The collection plan binds commit and lockfile; exp022 cells bind the manifest hash; downstream numbers bind campaign, commit, lockfile, manifest, dependencies, and training-run ID.
4. **Stochastic operations.** All reviewed runner-local NumPy and PyTorch stochastic operations use explicit constants, cell seeds, or stable derived seeds. Training seeds Python, NumPy, and PyTorch before model and loader construction.
5. **Dataset split symmetry.** Every exp022 cell uses the same seed-42 subsample and stratified 80/20 split; COBA and PING therefore see identical image identities within a scale.
6. **Collection output isolation.** The scheduler supplies distinct absolute state, derived, and log roots and `PINGLAB_REQUIRE_ISOLATED=1`; helpers reject partial or repository-artifact destinations.

Accepted limitations

1. No production checkpoints or outputs existed, so validation was assessed against manifest construction, validator behavior, focused synthetic tests, and current committed historical artifacts rather than a completed 90-cell bank.
2. The pooled-MNIST internal holdout remains the campaign evaluation partition; the audit requires accurate naming and counts but does not redesign the registered dataset split.
3. The dry planner requires a working uv executable even when it creates no scheduler jobs.

Validation

Focused campaign, argument, isolation, and smoke-cap tests reported 101 passed. The exact command is retained in `numbers.json`. The audit runner rechecked the captured commit, clean-manifest flag, production profile, 90 cells, 23 jobs in the dry plan, JSON validity, and a fail-closed sensitive-data scan. A complete Demolab build validates this article and its artifact references. No production training, Wilkes3 submission, RunPod job, Modal job, or other paid compute was run.

Conclusion

The collection graph and most reproducibility boundaries are coherent, but the campaign must not be frozen for production while #74 and #75 remain unresolved. Issues #76, #77, #78, and #79 must also be resolved or explicitly accepted before publication. The internal pooled-MNIST split should be described as checkpoint-selection holdout data, with its optimizer-training and holdout counts stated separately. Once the two blockers and the important follow-ups have evidence-backed dispositions, this audit can be rerun against the new frozen commit and the production readiness decision reconsidered.